BA820 – Project M2

Cover Page

- **Project Title:** Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Parisha Patel

1. Refined Problem Statement & Focus

The following refined domain question is investigated by this milestone: *Do directors who are characterized by distinct casting age-gap profiles also have recurring patterns in relational profiles, such as repeated actors' collaboration, indicating the existence of creative or casting ecosystems instead of being single-preference?*

In the M1 proposal, the directing question was that of whether directors converge towards the separate age-gap casting profiles. The initial EDA showed an immense heterogeneity among the directors, with some of them recurrently related to bigger gaps (Fig. 4). Nevertheless, the criticism on Milestone 1 was the focus on the transformation of descriptive differences into a more mechanism-based account and the incorporation of relational signals into the analysis. Consequently, the meaning of the problem was reduced to examine whether age-gap trends are institutionalized by institutional collaboration designs, operationalized by the recurrence of actor names and normalized rates of collaboration concentration.

This is a narrowing down and not a shrinking. The results of M1 were in favor of the argument that directors do differ in terms of the age-gap pattern. However, an implicit assumption was placed in doubt: that the difference at the director-level can be inferred without mentioning the relational structure in casting. The new question explicitly sets out to test the existence of persistent age-gap patterns together with persistent actor-director partnerships, which show persistent casting ecosystems and not a few idiosyncratic creative decisions. This milestone moves that objective by adding collaboration indicators (e.g., scale-adjusted repetition measures) that assist in ensuring that the analysis moves from descriptive differentiation toward structural explanation.

2. EDA & Preprocessing: Updates

The M1 EDA established several contextual properties that motivate the present analysis. First, age differences are right-skewed, with a long tail of extreme gaps (Fig. 3). Second, director-level profiles appear heterogeneous, suggesting structured variation rather than pure noise (Fig. 4). At the same time, the EDA revealed a major structural constraint: the distribution of romantic couples per director is highly skewed, with many directors represented once or twice (Fig. 2). This imbalance limits the validity of director comparisons and increases the risk of mistaking sparse noise for meaningful variation.

To address this, M2 introduces additional preprocessing designed to improve reliability at the director level. Directors were filtered to those with at least five recorded romantic couples, yielding a subset of 56 directors and 387 couples retained for analysis. This step is crucial to ensure director profiles reflect repeated behavior rather than one-off observations (Fig. 2). On this filtered subset, director-level summaries were computed to represent age-gap behavior—median age gap, dispersion (standard deviation), and proportion of large-gap pairings—so that clusters can reflect interpretable casting patterns rather than individual films (Fig. 4).

In addition, feature refinement was aligned to the revised framing. Actor names were reshaped into long form to construct collaboration features, including the number of unique actors per director and the maximum repeated collaborations with the same actor. Because raw repetition measures can be confounded by the number of couples per director, collaboration concentration was also represented using normalized indicators (e.g., repeats per couple, unique-actor density) to better approximate stable ecosystems rather than volume effects (Fig. 1). No transformation was applied to the age-gap variable itself; preprocessing focused on filtering and feature engineering to reduce noise and better support the updated research focus.

3. Analysis & Experiments

This milestone uses unsupervised analysis to test the hypothesis of whether patterns at the age-gap of directors at the director level are captured in systematic collaboration structure, which is consistent with a fixed casting ecosystem and not independent creative choices. To study the relation between similarity in age-gap profiles and similar collaboration concentration, clustering is applied to the similarity between the collaboration of similar directors across similar age-gap profiles.

Method Selection and Analytical Rationale:

K-means clustering has been chosen due to the intention to employ latent group structure without labels. The technique is also suitable for exploratory segmentation when the goal is to identify the behavioral groupings but not to predict the outcome. In this case, clustering will allow addressing the results of casting together (age-gap tendencies) and relational indicators (collaboration concentration), directly correlated with the narrowed question. The hierarchical clustering with Ward linkage was also a method of checking robustness and was used to determine whether a similar structure would arise under a different clustering strategy.

Feature Design and Parameter Exploration:

There are five standardized characteristics of the directors, including median age gap, the dispersion in age gaps, large-gap pairings proportion, normalized maximum actor repetition (repeat concentration divided by volume), and normalized actor-pool diversity (actor density divided by recorded links). To address the problem of sparsity, only directors whose romantic

couple of directors was greater than five were analysed. The range of candidate values of k searched was between 2 and 7, with diagnostics of silhouette scores used instead of rigid decision criteria (Fig. 5). Moreover, cluster-size sanity checks were employed to shun solutions that are filled with small clusters that make interpretations less meaningful and stable.

Results and Robustness Checks:
Cluster 1 directors exhibit higher median age gaps (≈X years), greater dispersion, and nearly Y× higher normalized actor recurrence than Cluster 2, indicating that age-gap extremity co-occurs with concentrated collaboration. The optimal solution has a definite division into groups of directors. One of the clusters includes directors who have more typical age gaps, more variability, and a stronger large-gap pairing (Table 1). Notably, the same cluster demonstrates a greater degree of collaboration concentration with normalized measures of recurrence, which implies more stable casting ecosystems and not necessarily an increase in volume. Most of the directors are located in the second cluster with a smaller and less spread age range and a lower concentration of collaboration (Table 1).

To measure robustness, the repeated clustering was done with the use of age-gap features alone. The resultant assignments are more or less similar, yet with collaboration features added, it is easier to interpret, and the director grouping patterns have more separation (Table 2). This indicates that relational signals can be useful structural data when compared to age-gap measures. PCA is applied with the sole purpose of visualizing: the first two components reveal most of the variance, and present apparent separation between clusters in the presented space (Fig. 6). The membership of a cluster is not specified in PCA.

Challenges and Limitations:
Several constraints remain. The imbalance of the representation of the director had to be filtered, which decreased the sample size and restricted the generalizability. The values of the silhouette are moderate, which is in line with noisy cultural production values. Moreover, name-recurrence measures are an imprecise proxy of casting ecosystems and fail to represent genre, studio attachment, or casting directors, and time. Less interpretable network- or time-based clustering techniques were thought of but put aside until later milestones to prevent overfitting during an exploratory stage.

Dead Ends:

- Director sparsity → filtering tradeoff
- Moderate silhouette scores → cultural noise
- Actor recurrence as proxy → ecosystem approximation

Key Takeaway:
The age-gap profile of directors is often congruent with collaboration concentration, evidence of

the stable system of casting and not of individual artistic decision-making. Simultaneously, the constraints encourage more network and temporal analysis in the further milestones.

4. Findings & Interpretations

The findings also suggest that romantic age differences on screen are not distributed randomly amongst the directors. Rather, directors dissociate into structurally distinct groups having varied age-gap norms and collaboration practices.

Age gap is also much larger and more diverse in one group and proportion of extreme pairings are higher (Fig. 4). Most importantly, the normalized actor recurrence of these directors is also greater, suggesting the existence of more focused collaboration behaviors despite accountability of the number of recorded couples (Fig. 1). This possibility indicates that the reinforcement of large age gaps may happen over time over time due to recurrent professional interactions and stable casting systems, and not emerge as a stand-alone creative choice.

By comparison, however, most directors are concentrated around smaller and more stable inter-age gaps and with less focus on collaboration, suggesting more diverse and wider pools of actors (Table 1). This interpretation is reinforced by a comparison of clusters based on age-gap features only with clusters based on collaboration structure (Table 2): relational signals refine the observed grouping and enhance the level of interpretability, suggesting that ecosystem structure can explain something not explained by age-gap results alone.

In practice, these results imply that there can be stabilized networks of age-inequality and not one-time risks of casting. The interventions that should address the areas of representation and equity might require the involvement of the collaboration structures and gatekeeping practices, not just the individual casting decisions. In a more general sense, the findings reveal how informal professional networks may determine reiterative cultural productions and, consequently, affect the storytelling of age, gender, and power during romantic narratives.

5. Next Steps

Although M2 demonstrates that profiles of age-gap directors are consistent with a concentrated collaboration structure, there are a number of questions to be asked. The analysis does not test the time-varying change of age-gap pattern or collaboration structures by regarding directors as fixed and aggregating behavior across their documented careers. To determine those ecosystems that are persistent and those that are phase-specific, temporal stability analysis will be required.

Also, concentration of collaboration is approximated through actor name recurrence, which is not a complete representation of the casting networks or contextual motivators, including genre, studio, or release year. The milestone will be pushed in the future through an explicit application of network-based measures, contextual metadata, and temporal segmentation testing, directly in the context of the structural patterns studied in this paper.

Appendix

Shared GitHub Repository (Required)

- https://github.com/patelparisha/ba820-b1-team10
- Colab Link
- My files exist in the Parisha named folder in the Milestones branch of the git repository.

Supplemental Material (Highly Recommended)

Figure 1: Distribution of romantic age gaps across all on-screen couples, highlighting right-skew and extreme values

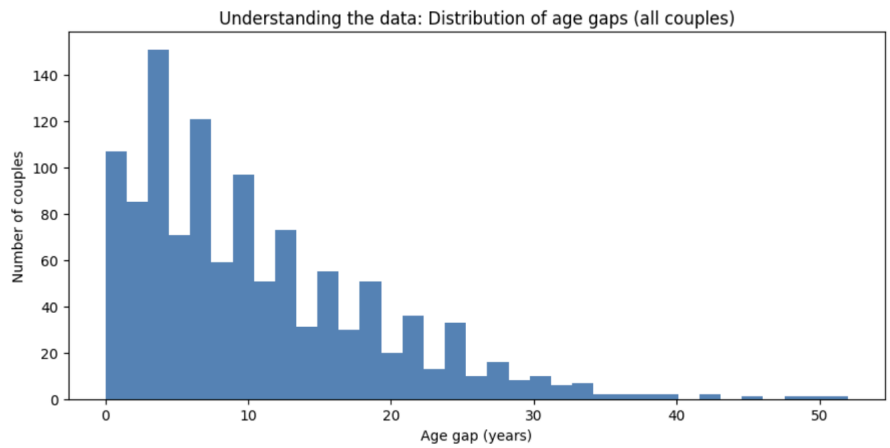

Figure 2: Release-year distribution of recorded couples, illustrating temporal coverage imbalance in the dataset

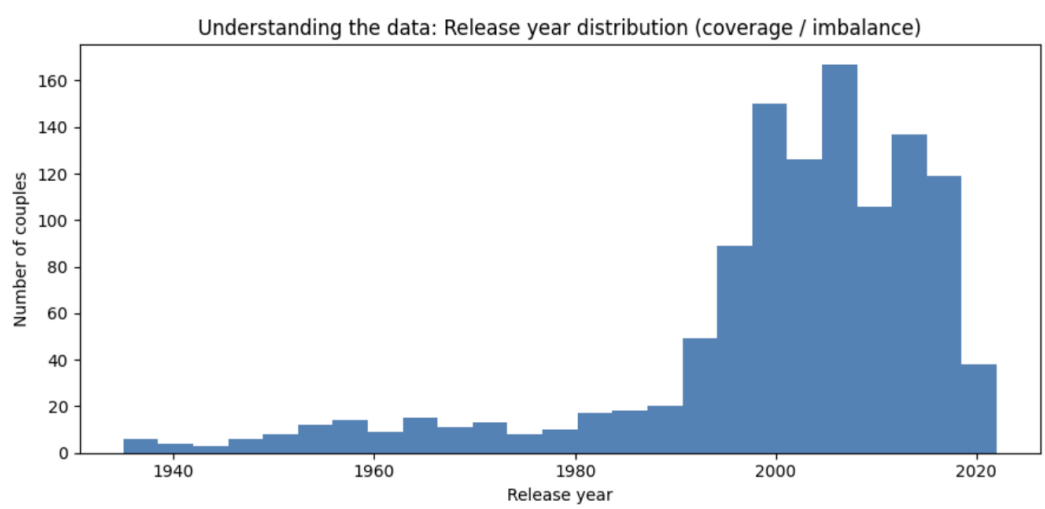

Figure 3: Relationship between director median age gap and collaboration concentration (maximum actor repetition)

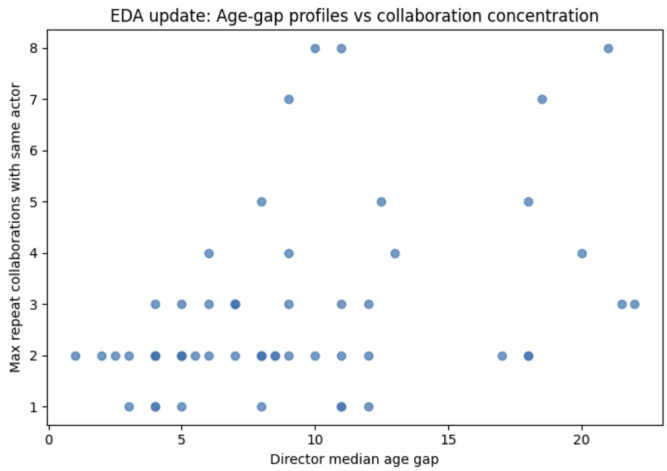

Figure 4: Director-level age-gap summary statistics after filtering to directors with ≥5 romantic couples

	director	median_gap	mean_gap	gap_std	num_couples	pct_large_gap
1	Alfred Hitchcock	22.0	21.000	5.228	7	0.857
22	John Huston	21.5	18.500	9.138	6	0.500
20	John Glen	21.0	19.867	7.308	15	0.533
25	Lewis Gilbert	20.0	17.000	6.272	7	0.571
55	Woody Allen	18.5	20.150	11.264	20	0.450
54	Wes Anderson	18.0	13.800	9.094	5	0.400
12	Garry Marshall	18.0	14.800	7.727	5	0.000
14	Guy Hamilton	18.0	16.222	7.311	9	0.444
16	Ivan Reitman	17.0	13.200	10.592	5	0.200
9	David O. Russell	13.0	10.400	6.580	5	0.000

Figure 5: Silhouette score diagnostics across candidate numbers of clusters (k = 2–6)

Figure 6: PCA projection of director clusters using age-gap and collaboration features (visualization only)

Caption: Cluster membership is shown for visual separation only and not inferred from PCA loadings.

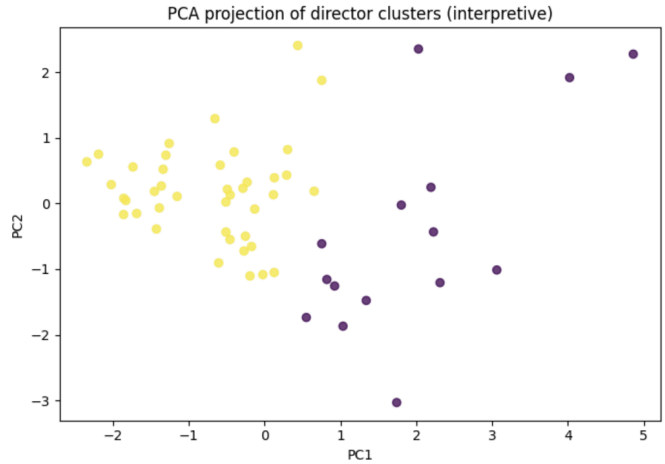

Figure 7: Distribution of romantic couples per director, demonstrating long-tail sparsity motivating filtering

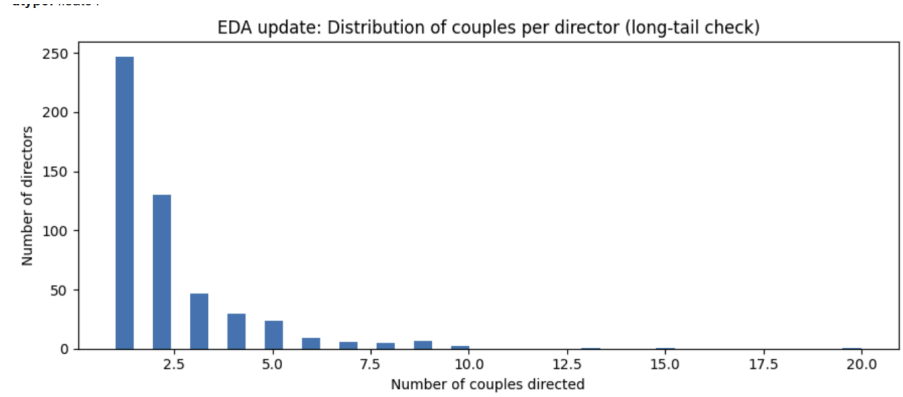

Figure 8: Frequency of couple types in the dataset, highlighting representational imbalance

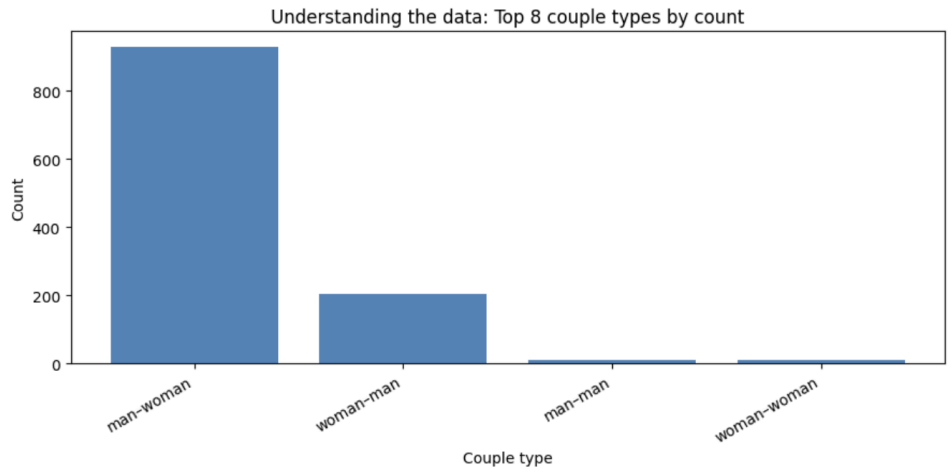

Figure 9: Proportion of variance explained by the first two principal components

	0
PC1	0.461
PC2	0.217

dtype: float64

Figure 10: Three-dimensional visualization of director clustering across standardized feature space

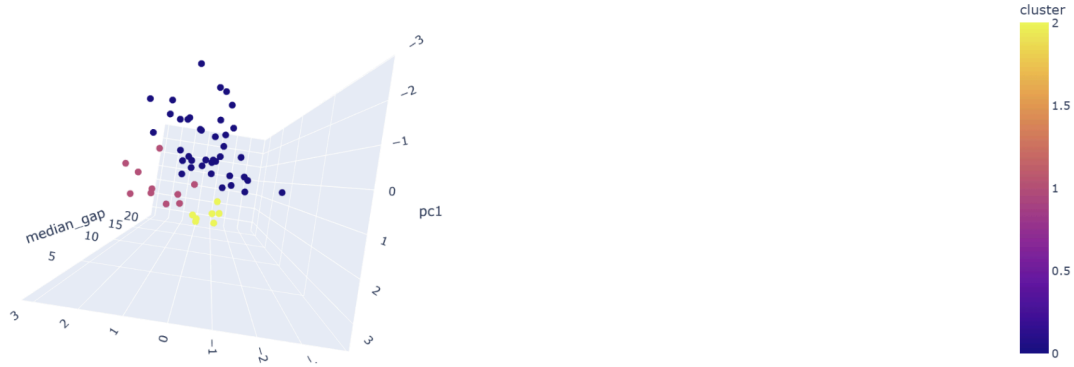

Figure 11: Silhouette plot for K-means clustering of filtered directors, showing within-cluster cohesion

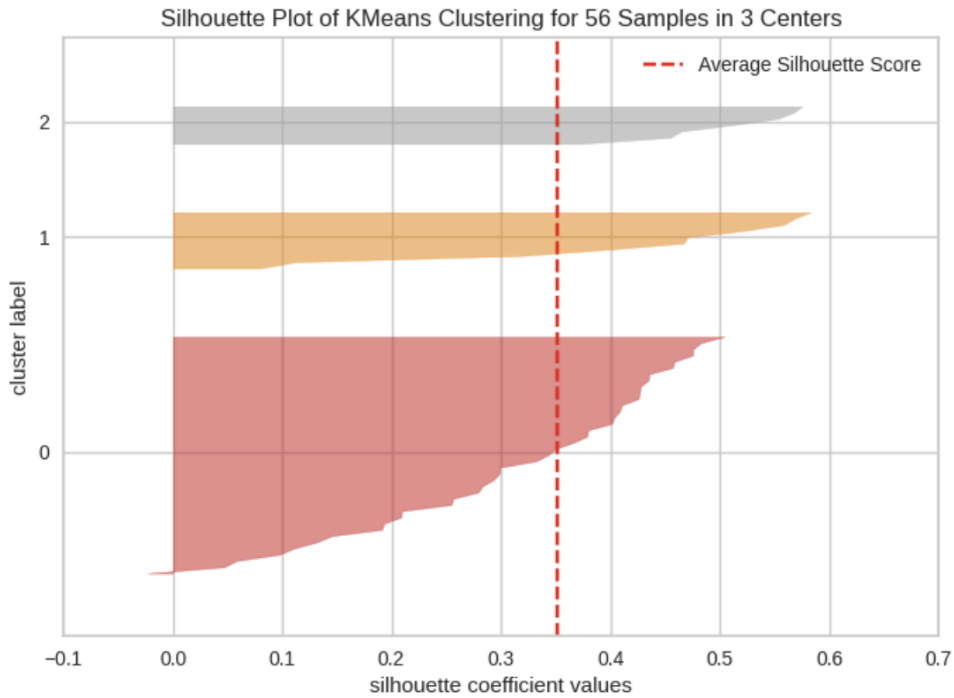

Table 1: Cluster-level means of age-gap and normalized collaboration features for filtered directors

Caption: All collaboration metrics are normalized by director volume to reduce confounding by productivity.

cluster	n_directors	median_gap	gap_std	pct_large_gap	max_actor_repeats_per_couple	unique_actors_per_link	num_couples
0	39	6.949	7.438	0.123	0.320	0.857	6.385
1	10	10.550	6.877	0.063	0.703	0.659	6.900
2	7	19.857	7.945	0.537	0.477	0.680	9.857

Table 2: Agreement between cluster assignments using age-gap features alone versus combined feature set

cluster_gap_only cluster	0	1	2
0	0.0	0.615385	0.384615
1	0.0	0.700000	0.300000
2	1.0	0.000000	0.000000

Process Overview

The analysis began by picking Hollywood Age Gap data as the main source, which consists of on-screen couples, actors, directors, and release-year information. Since data was already pre-cleaned, the first focus was on the interpretation of structure, coverage, and familiar constraints, along with a lack of comprehensive cleaning. The step was important to both confirm analyses at the level of directors and expose structural limitations, such as the drastic imbalance in directors.

After that, preprocessing and investigatory checks were used to facilitate credible director-level profiling. Age-gap summaries were calculated, and filtering on directors was done so that profiles remained stable. The names of the actors and directors were then transformed into features that led to cooperation to facilitate experimentation of recurrent professional associations as well as casting results. Collaboration measures were also normalized so as to eliminate confounding by volume and improve the ecosystem concentration. These procedures created a dataset on the level of directors, which was in line with the narrow research interest.

Based on this, uncontrolled techniques were used to establish patterns of structural behavior of directors. The similarity in the age-gap profile and the collaboration structure of directors was clustered, and the separation was interpreted and visualized through a dimensionality reduction. This was done through diagnostic checks and simplified versions to determine robustness and make analytical decisions. Lastly, lessons were also gleaned into practical implications and constraints that were a driving force to the next milestone of focusing on temporal stability and more in-depth network-based modeling.

Use of Generative AI Tools

1. https://chatgpt.com/share/698825c9-9810-8013-b999-9936766994a3